SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** 25038
- **Problem Statement Title -** Blockchain-Based Blue Carbon Registry and MRV System
- **Theme-** Clean & Green Technology
- **PS Category -** Software
- **Team ID -** MITADTSW138
- **Team Name (Registered on portal) :** CarbonX

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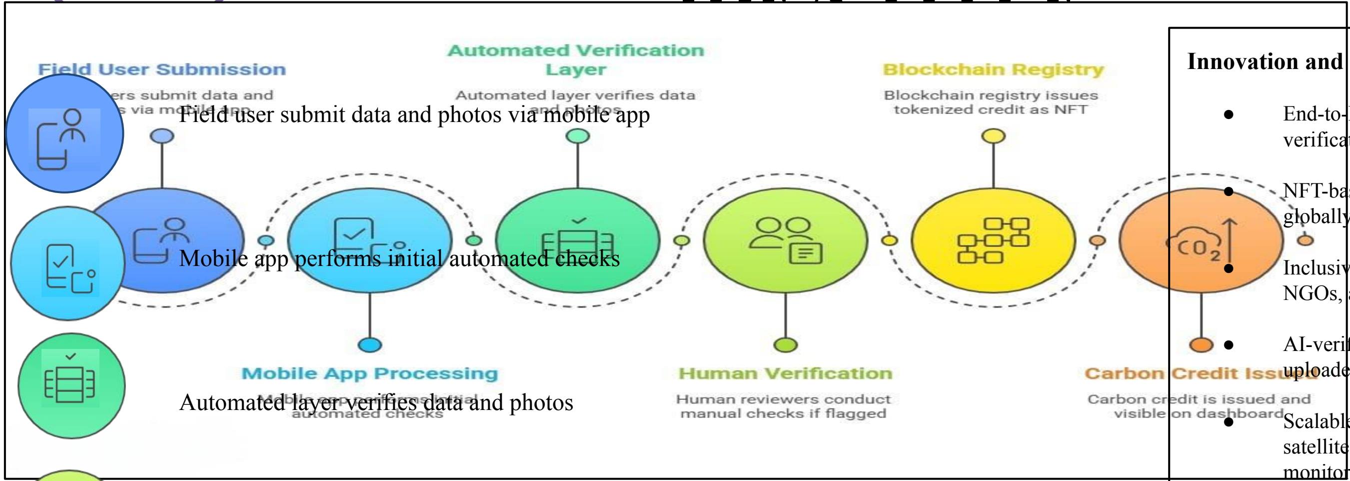
IDEA TITLE



Field user submit data and photos via mobile app

Mobile app performs initial automated checks

Automated layer verifies data and photos

Human reviewers conduct manual checks if flagged

Blockchain registry issues tokenized credit as NFT

Carbon credit is issued and visible on dashboard

Innovation and Uniqueness

- End-to-End MRV on Blockchain – from submission → verification → approval → tokenization.
- NFT-based Carbon Credits – verifiable, tradeable, and globally recognized.
- Inclusive by Design – empowers local Indian communities, NGOs, and panchayats to participate directly.
- AI-verified Photo Capture – prevents fraud by ensuring uploaded data's authenticity.
- Scalable Roadmap – future integration with drones, satellites, and IoT sensors for real-time ecosystem monitoring.

We propose a **Blockchain-based Blue Carbon MRV System** that ensures **authentic and transparent reporting** of coastal ecosystem restoration.

- **Mobile App** → Communities/NGOs upload **live geo-tagged photos** of plantations.
- **Verification** → AI + human checks prevent fake or AI-generated data.
- **Blockchain Registry** → Approved records are stored immutably, and **carbon credits are issued as NFTs** for traceability.

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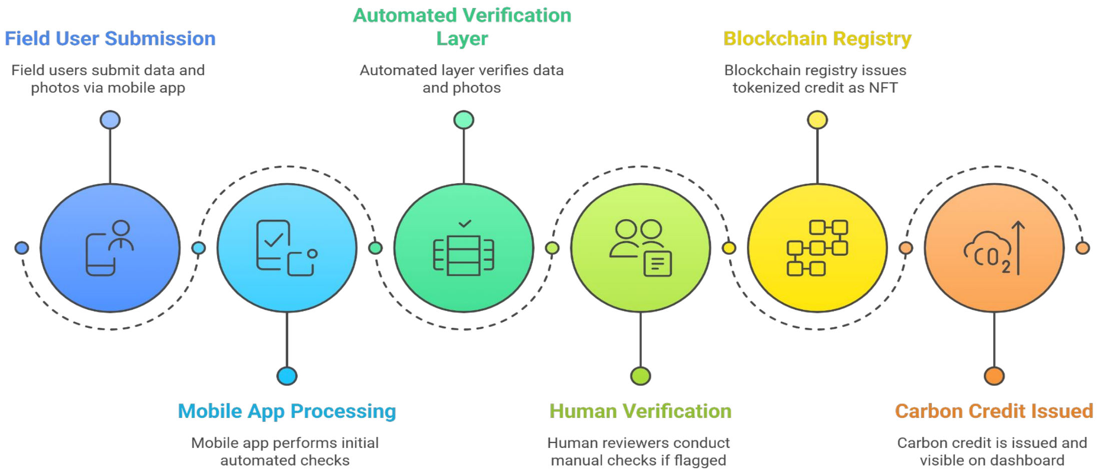

We propose a **Blockchain-based Blue Carbon MRV System** that ensures **authentic and transparent reporting** of coastal ecosystem restoration. It will include:

- **Mobile App** → Communities/NGOs upload live geo-tagged photos of plantations.
- **Geo-tagged live capture data + AI checks** → prevents fake/AI-generated submissions.
- **Blockchain Registry** → Approved records are stored immutably, and carbon credits are issued as NFTs for traceability.
- **Smart contracts** → automate tokenized carbon credit issuance.

How It Addresses the Problem

- **Blockchain ensures transparency** and prevents data tampering.
- **Verified field data** through geo-tagged photos and AI validation.
- **Streamlined MRV process** makes reporting faster and more reliable.
- **Smart contracts enable carbon credits**, creating tangible value for communities.

Innovation and Uniqueness

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TECHNICAL APPROACH

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- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

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FEASIBILITY AND VIABILITY

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S Feasibility

Technical Feasibility:

- Mobile app and web dashboard
- Blockchain integration
- AI-assisted photo verification

Operational Feasibility:

- NGOs, coastal panchayats, and government bodies are already stakeholders.
- User interface designed for simplicity and multilingual support ensures adoption.

W Potential System Challenges

Risks & Challenges:

- ➢ Fake or manipulated data submissions (old/gallery/AI-generated photos).
- ➢ Blockchain costs & transaction delays.
- ➢ Regulatory uncertainties around India's carbon credit markets.
- ➢ Limited technical capacity of grassroots users (NGOs, panchayats).
- ➢ AI photo verification accuracy not always reliable.

O Economic Sustainability Potential

Economic Feasibility & Potential:

- ❑ Low prototype cost.
- ❑ Long-term sustainability via carbon credit trading and partnerships.
- ❑ Creates a trusted carbon credit marketplace.
- ❑ Attracts green investments and CSR funding by ensuring verifiable, blockchain-backed credits.
- ❑ Scalable model with potential to expand into global carbon markets and ecosystem services.

T Overcome Challenges Strategies

- ★ For Fake Data, Enforce **live photo capture** and **AI-based authenticity check + final human review.**
- ★ For Blockchain issues, Use low-cost Polygon network and Batch transactions to reduce fees.
- ★ For Regulatory concerns, Align with NCCR & Indian CCTS guidelines.
- ★ For Technical Challenges, Provide a simple mobile UI with local language support. Training workshops for NGOs/panchayats.
- ★ For AI-related issues, We can use Hybrid verification.

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FEASIBILITY AND VIABILITY

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Feasibility Analysis

Economic Feasibilty

- Business Model(transaction fees) is **self-sustaining**.
- **Growing global and domestic demand** for verifiable carbon credits.

Operational Feasibilty

- Workflow **simplifies** the existing complex **MRV process**.
- **User-Friendly** mobile app for on-ground personnel.
- NCCR's role is enhanced, not replaced --> **Making adoption easier**

Technical Feasibility

- **Mature and well-documented technologies**(Blockchain,AI, Mobile Dev) are proposed.
- Architecture **Seperates heavy data(IPFS)** from the chain/ ledger --> Enhancing Performance

Potential Challenges and Risks	Strategies for Overcoming these Challenges
Fake or manipulated data submissions (old/gallery/AI-generated photos).	Enforce **live photo capture** and **AI-based authenticity check** + **final human review.**
Regulatory uncertainties around India's carbon credit markets.	Align with NCCR & Indian CCTS(Carbon Credit Trading Scheme) guidelines.
Limited technical capacity of grassroots users (NGOs, panchayats).	Provide a simple mobile UI. Training workshops for NGOs/panchayats.

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IMPACT AND BENEFITS

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Impact of the Solution:

Communities & NGOs

- Easy reporting of restoration work via mobile app.
- Builds trust in their contribution and ensures recognition.
- Reduces paperwork and manual follow-ups.

Government (NCCR)

- Faster verification & approval process.
- Reliable, tamper-proof data for policy-making.
- Ability to monitor progress across multiple regions in one dashboard.

Carbon Market Stakeholders

- Access to transparent, verified credits.
- Confidence in buying India-origin blue carbon credits.
- Lower risk of fraud due to blockchain traceability.

Technological & Innovation Impact

- First India-specific decentralized MRV solution for blue carbon.
- Showcases India's leadership in blockchain for climate innovations.

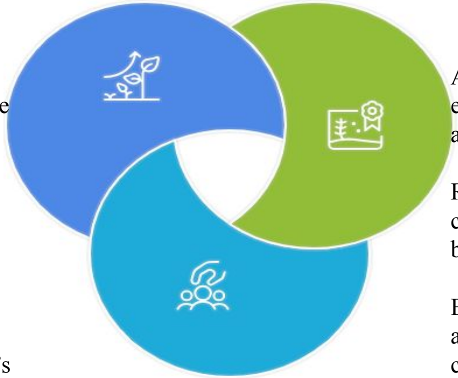
Benefits of the Solution:

Economical Benefits

Generates verifiable carbon credits tradable in domestic & international markets.

Opens new revenue streams for NGOs & communities.

Attracts foreign investment into India's climate projects



Environmental Benefits

Authentic MRV builds trust, ensuring real climate impact and global credibility.

Restores and protects blue carbon ecosystems, boosting biodiversity and marine life.

Enhances coastal resilience against floods, erosion, and climate change.

Social Benefits

Empowers local communities & fisherfolk with a trusted reporting platform.

Creates job opportunities in restoration & monitoring.

Builds community ownership of sustainability projects.

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IMPACT AND BENEFITS

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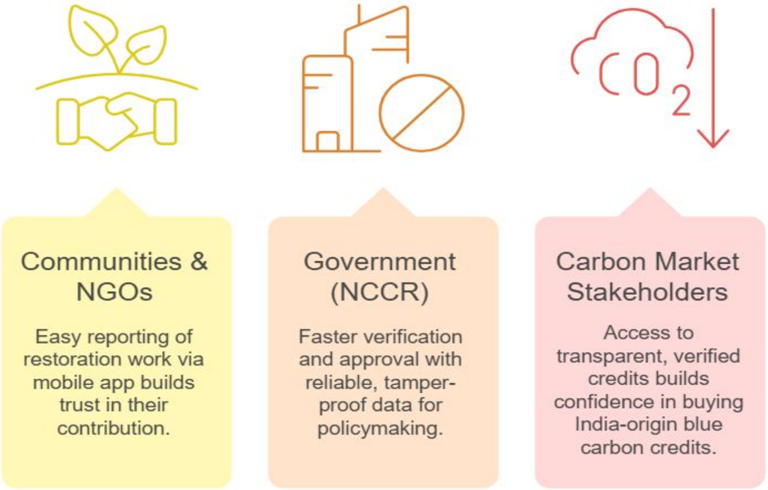
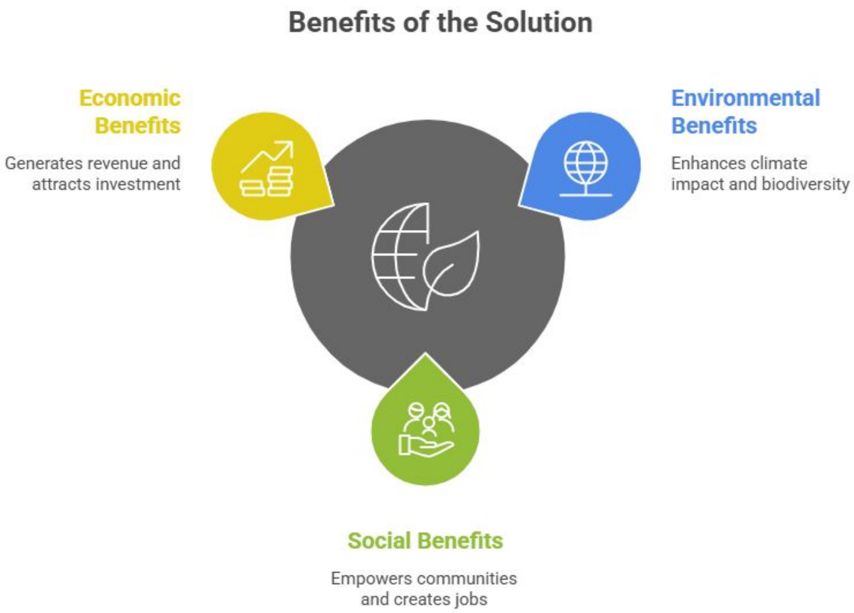
Potential impact on the target audience





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RESEARCH AND REFERENCES

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- https://timesofindia.indiatimes.com/city/kolkata/indias-carbon-market-potential-why-transparency-and-quality-are-key-to-eu-deals/articleshow/123256369.cms
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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.